Switchback AI Line
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An urban innovation operating system open to global AI agents

Centennial Jing-Zhang AI Innovation Belt international urban design call · conceptual proposal

SWITCHBACK AI LINE
SWITCHBACK AI LINE • AI-NATIVE URBAN OPEN PROTOCOL

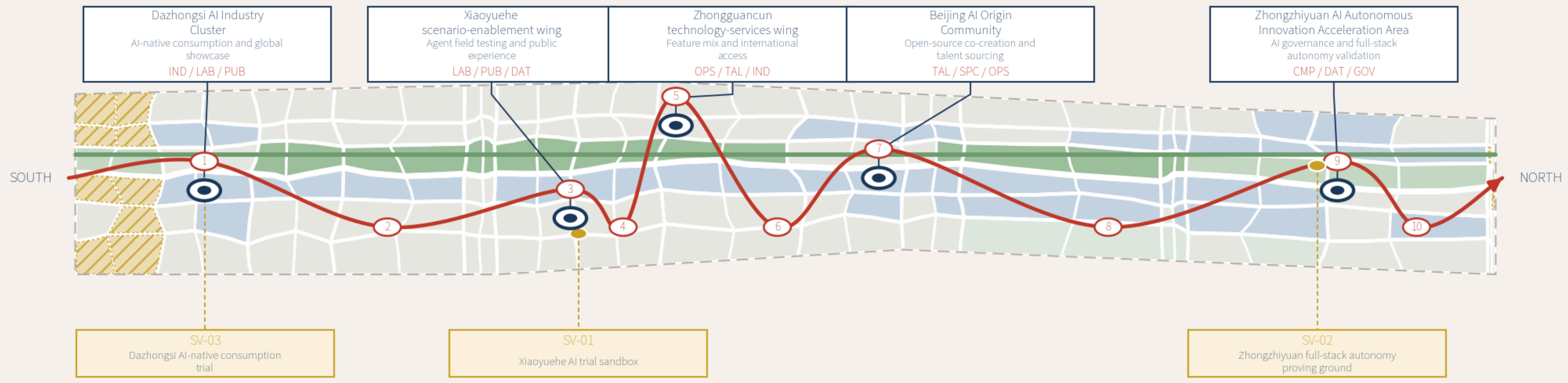
FIG-02
SITE OVERVIEW • OVERALL STRUCTURE AND SWITCHBACK ROUTE

SITE-WIDE SWITCHBACK STRUCTURE

A TEN-MOVE STRATEGIC FRAMEWORK

A railway that switched back to solve the impossible, upgraded into a city that global AI agents can call, verify and iterate against.

L1 Belt brand Switchback AI Line	L2 Operating mechanism Switchback Protocol	L3 Capability node Official district name + role node	L4 Event system Assembly / Sprint / Open Switchback	L5 Honours system Switchback Mark	L6 Scenario code SB-<interface>-<district>-<no.>



LEGEND

Features come from the package GeoJSON; values come from metrics.json

- **Innovation Fabric** — Research 0802 / commercial 05
- **Reserved Land** — Class 16 • 4.00%
- **Cultural Corridor** — 17 segments • continuous
- **Cultural Spine** — AIZ-LIN-01 continuous link
- **Capability Node** — 5 • matching three areas and two wings
- **Switchback Route** — 10 moves • switchback ascent
- **SV industry test and validation scenario** — 3 • each requires case-by-case human approval
- **Provisional study extent** — Not an official redline; pending verification

SWITCHBACK LOGIC

Constraints are not obstacles; they are the conditions that generate the route

CONSTRAINT
Provisional boundaries, data gaps, multi-party collaboration

TURN
Trade route length for gradient instead of forcing the climb;
trade mechanism for certainty instead of inventing data

VALIDATION
Controlled testing in real space,
with human abort available every time

ITERATION
Public comment and review findings feed back into the governance register
and shape the next admission round

UPGRADE
Consolidated into an open protocol other cities can cite

! MOBILITY AND SERVICE STRUCTURE IS CONCEPTUAL AND CARRIES NO ENGINEERING CONCLUSION.

Routes, nodes and service zones are conceptual. No conclusion is offered on road boundary lines, lanes, alignment, structures, underground space or rail alignment. Siting must be rechecked once official boundaries and regulatory controls exist.

DRAWING DATA	
ORIENTATION	N → north is right
CORRIDOR S–N / E–W	9.7 km / 1.33 km
NODES / SCENARIOS	4 + 1 / 10
BOUNDARY STATUS	PROVISIONAL
REV	A • 2026-08

SOURCE
[source:BOUNDARY-SOURCE] [source:KEY-AREA-SOURCE]
[source:DESIGN-SERVICE-ZONE] [source:DESIGN-LAND-USE]

DATA
ai_service_zone • scenario_node • land_use
green_space • roads • metrics.json

STATUS
Conceptual Design Proposal
not statutory planning • not approved

NOTE
All spatial boundaries are provisional and subject to official verification.

Overall structure and switchback route. Every feature comes from the package GeoJSON and every value from metrics.json.

> "Switchback" Named after the switchback at Qinglongqiao station on the Jing-Zhang railway. A switchback is not a detour; it accepts that the grade is real and keeps climbing another way. The proposal treats that move as a method: facing a densely built linear district with complicated tenure, it does not pretend the area can be reshaped in one gesture, but repairs the neglected urban fundamentals first and then layers on capabilities global AI agents can call, verify and iterate against.

Core trade-off: repair the fundamentals, then add capability

This proposal spends its effort on repairing urban fundamentals rather than layering on a smart-city narrative. A scheme with an inadequate street network and road land in the single digits does not meet the basic bar for urban design, whatever narrative sits on top of it.

9.31
Network density km/km²
Street-block requirement ≥ 8

15.09%
Road land share
National standard 10%–25%

29.24%
Residential land share
Housing-led built-up district

16.80%
Green space and plaza share
Held under 20%

1.40
Gross FAR
Back-calculated from recommended massing

533
Building count
3748 m average per building²

10
AI scenario node
10 interface types

15
Renewal projects
3 phase

Four counter-intuitive decisions

Push the green share down
Green reads well on a drawing, but in a district that is already fully occupied, Every percentage point of green share cancels some existing use. At a third of the site, a scheme is in effect proposing large-scale demolition without saying so.

Restore the housing share
housing is not stock to be cleared; it is the thing to be served. Without residents, AI scenarios have nothing to validate against.

Do not state a FAR directly
Official control conditions are unavailable, so the value is back-calculated from the recommended massing: Footprint times assumed storeys gives total building scale; dividing by the project extent gives the gross FAR. Both numbers come from one calculation and cannot contradict each other.

Put the flagship district last
Phasing is ordered by implementability, not by importance. Starting with a district that needs a regulatory-plan amendment would leave the scheme on paper.

Land use: from road-network partition to land-use assignment

The land-use partition is not block-by-block colouring over a base map. Road centrelines come first, road land is buffered out of them by hierarchy, and the remaining blocks are then assigned a use, so centrelines, road land and block edges stay strictly consistent.

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FIG-03
LAND USE STRUCTURE • LAND USE AND DEVELOPMENT INTENSITY

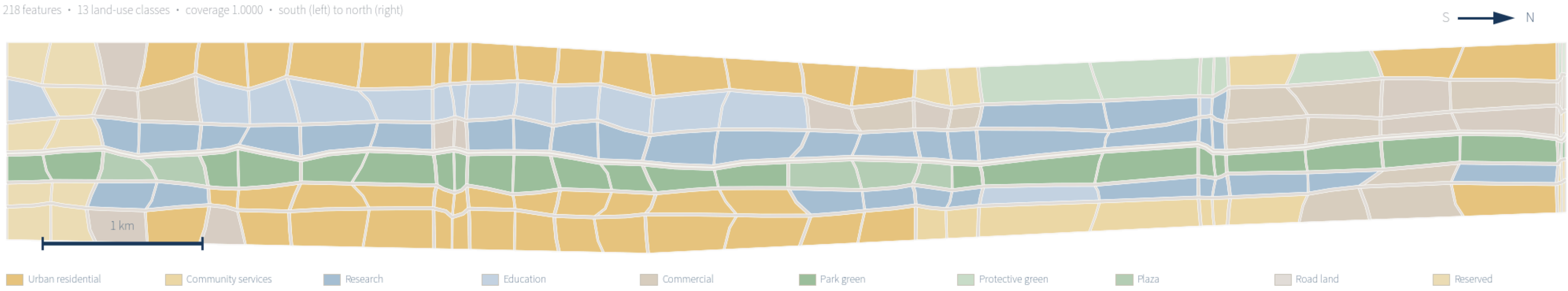
From road-network partition to land-use assignment

LAND USE DERIVED FROM THE STREET GRID, NOT PAINTED OVER IT

Road land is cut from the centrelines first, then blocks are assigned a use;
the partition covers the whole extent with no gaps, no overlaps, fully recomputable.

LAND USE PLAN

218 features • 13 land-use classes • coverage 1.0000 • south (left) to north (right)



COMPOSITION

Share of project extent; values come from metrics.json and match the GeoJSON recomputation

Urban housing + community services 0701/0702 — 29.24%
Restored to the realistic level of a housing-led built-up district

Urban road land 1207 — 15.09%
GB 50137 requires 10%–25%

Research land 0802 — 15.01%
Matching universities, institutes and innovation-transfer space

Green space and plazas 1401/1402/1403 — 16.80%
Held under 20% to avoid implicit demolition

Reserved land 16 — 4.00%
Reserved for AI activities not yet defined

INTENSITY

Official regulatory controls are unavailable; the values below are recommendations

Gross FAR — 1.40
Footprint × assumed 8 storeys ÷ project extent

Total building scale — 1598×10^4 m^2
Back-calculated from the same source as the FAR

Building footprint — 200×10^4 m^2
Share of extent 17.5%

Building count — 533 buildings
Single-building grain, not block-sized

Mean footprint per building — 3748 m^2
about 0.37 ha

RETAIN / RENEW

Housing, education, health and sports are never listed for demolition review

retain — 338 buildings

renew — 170 buildings

future — 13 buildings

assess — 12 buildings

assess recommends opening a low-efficiency space review. It is not a settled demolition decision. Housing, education, health and sports buildings never enter this category.

SOURCE
[source:DESIGN-LAND-USE] [source:LAND-USE-CLASSIFICATION] [standard:MNR-LAND-USE-CLASSIFICATION-GUIDE]

DATA
land_use • buildings • roads
metrics.json

STATUS
Conceptual Design Proposal
not statutory planning • not approved

NOTE
All spatial boundaries are provisional and subject to official verification.

Land use and development intensity. 218 features, 13 land-use classes, coverage 1.0000, no overlaps and no gaps.

Building scale and grain

Building footprint	**200 ×10⁴ m²**
Total building scale	**1598 ×10⁴ m²**
Building count	**533 buildings**
Mean footprint per building	**3748 m²**

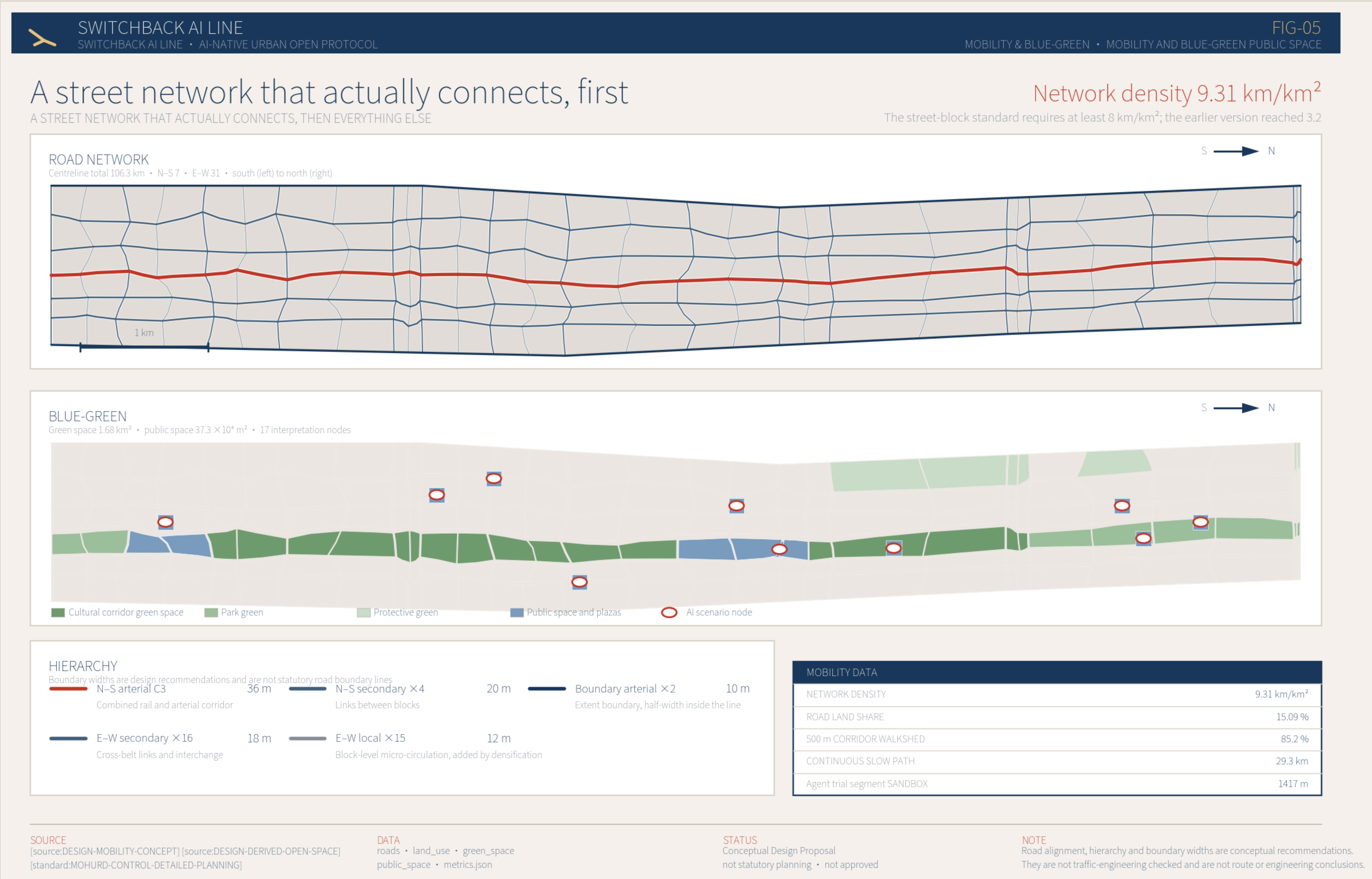
Massing at single-building grain gives daylight, spacing, fire access and frontage discussions something concrete to work on.

Retain / renew / rebuild

retain	**338 buildings**
renew	**170 buildings**
future	**13 buildings**
assess	**12 buildings**

Housing, education, health and sports are retained without exception. A recommendation to review is not a demolition decision.

Mobility: a street network that actually connects, first



Road system and blue-green public space. Road centrelines total 106.3 km, density 9.31 km/km²。

Knock-on effects of a denser street network

Block depth drops from about 640 m to about 320 m, so buildings can be cut to single-building grain and each block can become a renewal unit on its own; More intersections bring walking routes closer to straight-line distance, and a higher road-land share leaves far more room for utilities. The street network is a precondition for every other spatial decision, not a layer to be added last.

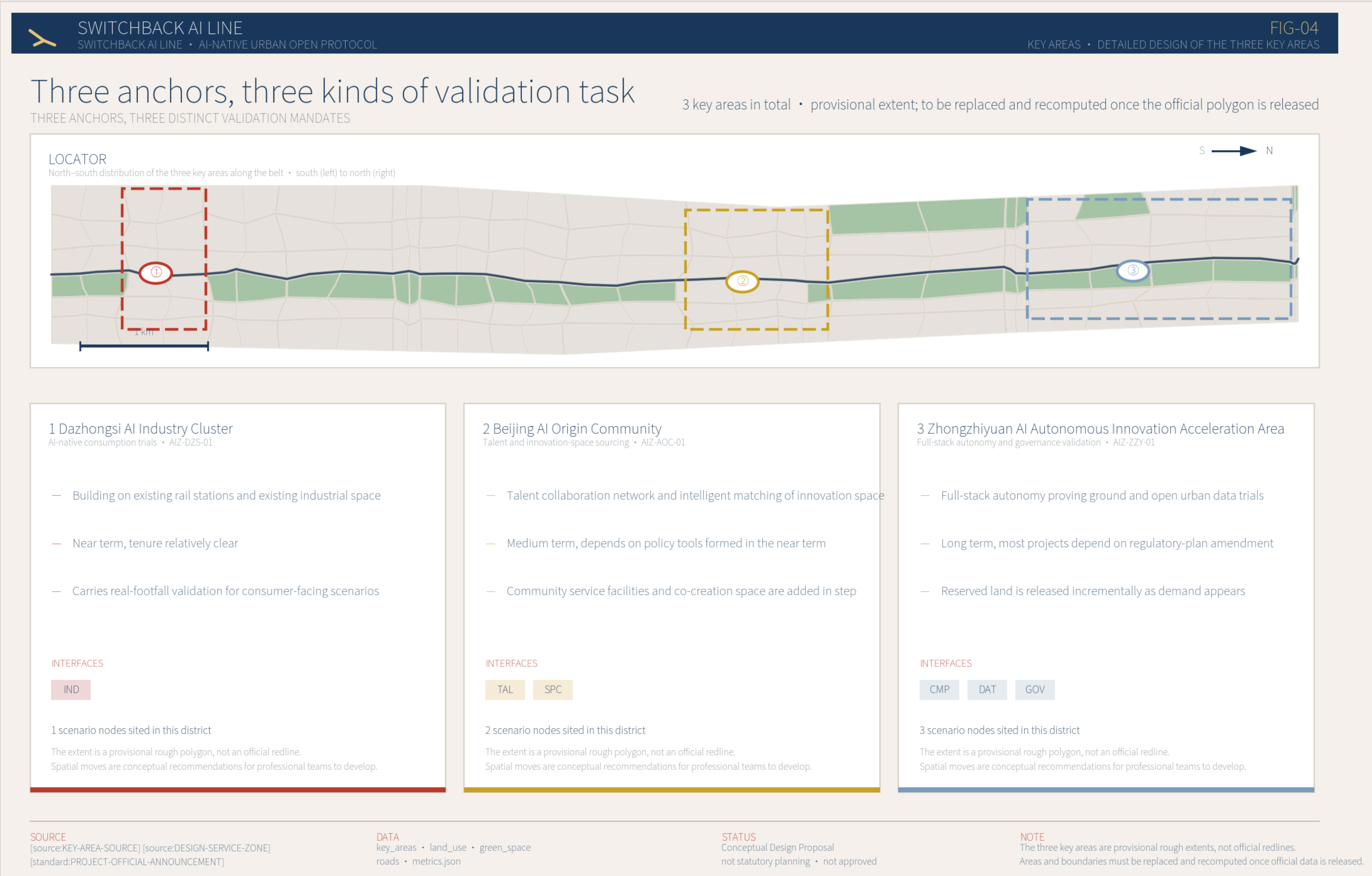
Blue-green and public space

Green space system	**1.68 km²**
Public space	**37.3 ×10⁴ m²**
Continuous slow path	**29.3 km**
Interpretation nodes	**17**

Public life rests on streets rather than plazas: once the network is denser, the street itself becomes the largest public space.

Three anchors, three kinds of validation task

The three key areas should not become three interchangeable AI parks. Together they form one validation chain: Dazhongsi tests whether consumers will use it, the Origin Community tests whether practitioners will stay, and Zhongzhiyuan tests whether the governance mechanism holds.



Dazhongsi AI Industry Cluster

Real-footfall validation of AI-native consumption

Next to an existing rail station, with mature retail and housing around it, this is the only one of the three where a consumer-facing scenario could be used by real people on day one. The moves are mainly frontage retrofits and opening local streets, which need no regulatory-plan amendment, so it comes first.

Beijing AI Origin Community

Sourcing of talent and innovation space

The distinctive resource here is people, not land: researchers, founders, students and long-term residents live at high density together. Small and medium co-creation spaces, rentable by the hour and reachable from the street, are added alongside community services so newcomers and existing residents share one set of everyday facilities.

Zhongzhiyuan AI Autonomous Innovation Acceleration Area

Full-stack autonomy and governance validation

The largest area, with the lowest existing intensity and the best reserve capacity, suits tasks that need space and time. Most moves depend on regulatory-plan amendment and land-use change, so it comes last — an honest acknowledgement of the policy cycle.

Ten scenario cards

SC-01	Xiaoyuehe AI trial sandbox	LAB
SC-02	Zhongzhiyuan full-stack autonomy proving ground	CMP
SC-03	Dazhongsi AI-native consumption trial	IND
SC-04	AI Talent collaboration network	TAL
SC-05	Intelligent matching of innovation space	SPC
SC-06	Open urban data trials	DAT
SC-07	Jing-Zhang AI cultural guide	CUL
SC-08	Public AI experience feedback loop	PUB
SC-09	Global AI innovation event network	OPS
SC-10	AI Governance validation register	GOV

Phasing: ordered by implementability

Period	Start condition	Delivery party
Near term 2026–2028 Dazhongsi and Zhongguancun services wing start-up phase	Work with clear tenure that can start without a regulatory-plan amendment	District platform company / title holders / market participants
Medium term 2029–2032 Beijing AI Origin Community formation phase	Depends on renewal experience and policy tools formed in the near term	District platform company / universities and institutes / private capital
Long term 2033–2035 Zhongzhiyuan full-stack autonomy and governance validation phase	Most projects depend on regulatory-plan amendment or new quotas	District platform company / market participants / long-term operator

Self-imposed implementability tests

Beyond the format validation, the proposal sets itself an additional group of tests derived from public standards. These rules are the agent's requirements on itself, not an official review conclusion. The first version failed five of them.

Test	Value	Requirement	Result
Urban road land share	**0.150922**	0.1 - 0.25	Pass
Network density km/km²	**9.31283**	>= 8.0	Pass
Park green space share	**0.108512**	<= 0.15	Pass
Combined green space and plaza share	**0.168003**	<= 0.2	Pass
Residential land share	**0.292449**	>= 0.25	Pass
Mean footprint per building m²	**3747.59**	<= 20000	Pass
Number of phases	**3**	>= 3	Pass
Land-use coverage	**1**	0.999 - 1.001	Pass
Maximum gap or overspill m²	**0**	<= 1000.0	Pass
Recommended FAR	**1.4002**	0.01 - 12.0	Pass
Unknown metrics missing a reason or resolution_state	**0**	<= 0	Pass

> The project extent and the three key areas both use provisional rough polygons inferred from public material, not official redlines. Once official data is released, land use, roads, green space, buildings, phasing and every metric must be recomputed as a whole. The package is produced by a reproducible build pipeline; the whole process above can be re-run.